Ransomware Attack Simulation

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Ransomware

Ransomware is a type of malware that prevents the user from accessing their device and data stored on it, by encrypting the users' files. Cybercriminals will use ransomware to hold data hostage, demanding payment in exchange for the decryption key. Phishing emails and malicious websites are various ways ransomware can spread (Kosinski, 2025).

Types of ransomwares

There are two types of ransomware called, encrypting ransomware also known as crypto ransomware. It holds the targets' data hostage by encrypting it then the attackers demands a ransom in exchange for the encryption key necessary to decrypt the data (Kosinski, 2025).

The other type is called non-encrypting ransomware or screen-locking ransomware. It locks the targets' entire device by blocking access to the operating system. The device displays a screen that makes the ransom demand instead of starting up (Kosinski, 2025).

There are subcategories to the two types of ransomware:

Leakware or doxware

Is ransomware that takes sensitive data and threatens to publish it (Kosinski, 2025).

Mobile ransomware

Is ransomware that includes all ransomware that affects mobile devices, which happens through malicious apps or drive-by downloads. Most mobile ransomware is non-encrypting ransomware. Because of automated cloud data backups hackers prefer screen-lockers for mobile attacks which makes it easy to reverse encryption attacks (Kosinski, 2025)

Wipers

Threatens to destroy data if the target decides not to the ransom, and even if the target pays the ransom the ransomware still destroys data. It is often deployed by nation-state actors or hacktivists rather than common cybercriminals (Kosinski, 2025).

Scareware

Is ransomware that intents to scare users into paying a ransom, scareware can pretend to be law enforcement and send out messages accusing the target of a crime while demanding a fine. On the other hand, it might spoof a legitimate virus infection alert by encouraging the target to purchase ransomware disguised as antivirus software (Kosinski, 2025).

Stages of a ransomware attack

A ransomware attack usually goes through these stages:

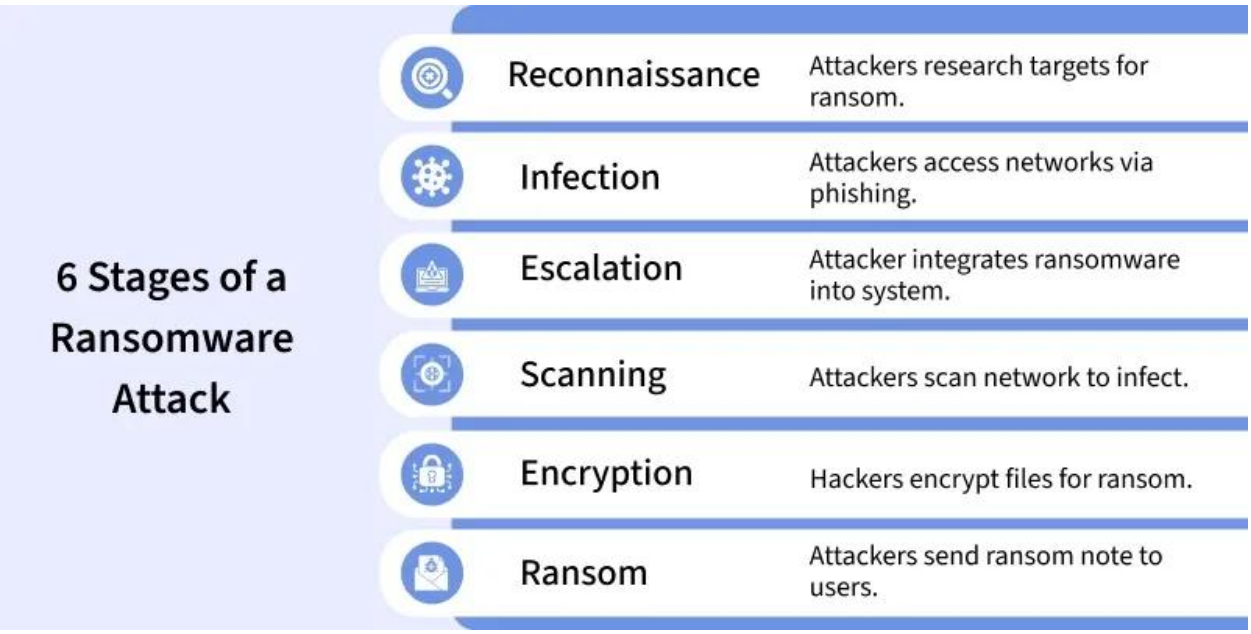


Figure 1: (geeksforgeeks, 2025)

How Ransomware works.

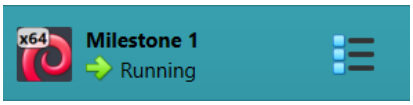
It run through a cycle before the targeted user is even fully aware that they may have been diagnosed with malware infection.

- Infection: targeted users install ransomware without any knowledge because the primary attack vector occurs through phishing emails, drive-by downloads, links and comprised software (geeksforgeeks, 2025).
- Execution: with nearly uncrackable security encryption after installation the program delivers a payload that searches for files with sensitive data only to encrypt those files (geeksforgeeks, 2025).
- Encryption: the targeted user will receive a message that the attacker wants ransom in exchange for the decryption key that only the attacker has access to (geeksforgeeks, 2025)
- Ransom Demand: bitcoins is the usual currency the attackers choose as ransom (geeksforgeeks, 2025).
- Decryption: the decryption key is sent only if the targeted user pays the ransom but data retrieval most of the time may not be recovered (geeksforgeeks, 2025).

The step-by-step process on how to execute a simulated ransomware attack

Step 1: Environment Setup and Payload script creation.

Proof of Virtual Machine, Milestone 1.



Create a new file named Milestone_1_payload.py in home directory using Mousepad in the kali terminal.

File Edit Search View Document Help

```python
#!/usr/bin/env python3
import os            # Command used for system commands like logout
import time          # Command used for the 30-second delay
import webbrowser    # Command used to open 9gag.com
from tkinter import messagebox, Tk # Used for the ransomware popups

#—— 1. Create Payload File ——

def create_target_file():
    # Below are the 5 paragraphs of text from lipsum.com
    content = """[ What is Lorem Ipsum?
Lorem Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s, when an unknown printer took a galley of type and scrambled it to make a type specimen book. It has survived not only five centuries, but also the leap into electronic typesetting, remaining essentially unchanged. It was popularised in the 1960s with the release of Letraset sheets containing Lorem Ipsum passages, and more recently with desktop publishing software like Aldus PageMaker including versions of Lorem Ipsum.

Why do we use it?
It is a long established fact that a reader will be distracted by the readable content of a page when looking at its layout. The point of using Lorem Ipsum is that it has a more-or-less normal distribution of letters, as opposed to using 'Content here, content here', making it look like readable English. Many desktop publishing packages and web page editors now use Lorem Ipsum as their default model text, and a search for 'lorem ipsum' will uncover many web sites still in their infancy. Various versions have evolved over the years, sometimes by accident, sometimes on purpose (injected humour and the like).

Where does it come from?
Contrary to popular belief, Lorem Ipsum is not simply random text. It has roots in a piece of classical Latin literature from 45 BC, making it over 2000 years old. Richard McClintock, a Latin professor at Hampden-Sydney College in Virginia, looked up one of the more obscure Latin words, consectetur, from a Lorem Ipsum passage, and going through the cites of the word in classical literature, discovered the undoubtable source. Lorem Ipsum comes from sections 1.10.32 and 1.10.33 of "de Finibus Bonorum et Malorum" (The Extremes of Good and Evil) by Cicero, written in 45 BC. This book is a treatise on the theory of ethics, very popular during the Renaissance. The first line of Lorem Ipsum, "Lorem ipsum dolor sit amet..", comes from a line in section 1.10.32.

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```

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    with open("Milestone_1_Payload.txt", "w") as f:
        f.write(content)

# ——2. Scramble and Password Protect (Encryption) ——

def encrypt_file():
    # Read the original text
    with open("Milestone_1_Payload.txt", "r") as f:
        original_text = f.read()

    # Scramble: Reverse the text (Reversible by running again)
    scrambled = original_text[::-1]

    with open("Milestone_1_Payload.txt", "w") as f:
        f.write(scrambled)

    # Password protect into a ZIP using the system's zip tool
    # Password is : Wednesd@y2025
    os.system('zip -P Wednesd@y2025 Milestone_1_Payload.zip Milestone_1_Payload.txt')

    # Remove the unzipped file to simulate 'locking' the user out
    os.remove("Milestone_1_Payload.txt")

# —— 3. Trigger popups and web link ——

def show_warnings():
    root = Tk()
    root.withdraw() # Hide the main tiny Tkinter window

    # Realistic Ransomware Popup
```

```python
# —— 3. Trigger popups and web link ——

def show_warnings():
    root = Tk()
    root.withdraw() # Hide the main tiny Tkinter window

    # Realistic Ransomware Popup
    messagebox.showerror("!!! SYSTEM ENCRYPTED !!!", "Your personal files are now scrambled. Send 0.1 BTC to recover access.")

    # Malicious Program Popup
    messagebox.showwarning("Warning", "Backdoor established. System logs are being uploaded...")

    # Open 9gag.com
    webbrowser.open("https://9gag.com/")
    root.destroy()

# —— 4. System logout ——

def final_countdown():
    # Wait for 30 seconds as required
    time.sleep(30)
    # Force logout for Kali Linux
    os.system("pkill -u $(whoami)")


#—— Main Execution ——

if __name__ == "__main__":
    create_target_file()
    encrypt_file()
    show_warnings()    # This function opens the popups
    final_countdown()  # This function waits 30s THEN logs out
```

The actual code:

```
#!/usr/bin/env python3

import os          # Command used for system commands like logout

import time        # Command used for the 30-second delay

import webbrowser   # Command used to open 9gag.com

from tkinter import messagebox, Tk # Used for the ransomware popups


#--- 1. Create Payload File ---


#def create_target_file():

  # Below are the 5 paragraphs of text from lipsum.com

  content = """[ What is Lorem Ipsum?
```

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```
to generate Lorem Ipsum which looks reasonable. The generated Lorem Ipsum is therefore always
free from repetition, injected humour, or non-characteristic words etc. ]"""
  with open("Milestone_1_Payload.txt", "w") as f:
    f.write(content)

# ---2. Scramble and Password Protect (Encryption) ---

#def encrypt_file():
  # Read the original text
  with open("Milestone_1_Payload.txt", "r") as f:
    original_text = f.read()

  # Scramble: Reverse the text (Reversible by running again)
  scrambled = original_text[::-1]

  with open("Milestone_1_Payload.txt", "w") as f:
    f.write(scrambled)

  # Password protect into a ZIP using the system's zip tool
  # Password is : Wednesd@y2025
  os.system('zip -P Wednesd@y2025 Milestone_1_Payload.zip Milestone_1_Payload.txt')

  # Remove the unzipped file to simulate 'locking' the user out
  os.remove("Milestone_1_Payload.txt")

# --- 3. Trigger popups and web link ---

#def show_warnings():
  root = Tk()
```

```
    root.withdraw() # Hide the main tiny Tkinter window

    # Realistic Ransomware Popup
    messagebox.showerror("!!! SYSTEM ENCRYPTED !!!", "Your personal files are now scrambled. Send 0.1 BTC to recover access.")

    # Malicious Program Popup
    messagebox.showwarning("Warning", "Backdoor established. System logs are being uploaded...")

    # Open 9gag.com
    webbrowser.open("https://9gag.com/")
    root.destroy()

# --- 4. System logout ---

#def final_countdown():
    # Wait for 30 seconds as required
    time.sleep(30)
    # Force logout for Kali Linux
    os.system("pkill -u $(whoami)")

#--- Main Execution ---

#if __name__ == "__main__":
    create_target_file()
    encrypt_file()
    show_warnings()       # This function opens the popups
   # final_countdown()  # This function waits 30s THEN logs out
```

The explanation of the code:

Step1: The Import Block

It is important to import the necessary modules at the top of your script.

```
#!/usr/bin/env python3
import os          # Command used for system commands like logout
import time        # Command used for the 30-second delay
import webbrowser  # Command used to open 9gag.com
from tkinter import messagebox, Tk # Used for the ransomware popups
```

Step2: Creation of the target file

Define a function to generate the text file using Lorem Ipsum text.

```
#--- 1. Create Payload File ---

#def create_target_file():
  # Below are the 5 paragraphs of text from lipsum.com
  content = """[ What is Lorem Ipsum?
```

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```
with open("Milestone_1_Payload.txt", "w") as f:
    f.write(content)
```

Step3: Scramble and Password Protect

Here we have to make the “encryption” reversible which we will have to reverse the string. Then we Linux zip will be used to add a password.

```
# ---2. Scramble and Password Protect (Encryption) ---


#def encrypt_file():
  # Read the original text
  with open("Milestone_1_Payload.txt", "r") as f:
```

```
    original_text = f.read()

  # Scramble: Reverse the text (Reversible by running again)
  scrambled = original_text[::-1]

  with open("Milestone_1_Payload.txt", "w") as f:
    f.write(scrambled)

  # Password protect into a ZIP using the system's zip tool
  # Password is : Wednesd@y2025
  os.system('zip -P Wednesd@y2025 Milestone_1_Payload.zip Milestone_1_Payload.txt')

  # Remove the unzipped file to simulate 'locking' the user out
  os.remove("Milestone_1_Payload.txt")
```

Step4: The UI and Website redirect

```
# --- 3. Trigger popups and web link ---

#def show_warnings():
  root = Tk()
  root.withdraw() # Hide the main tiny Tkinter window

  # Realistic Ransomware Popup
  messagebox.showerror("!!! SYSTEM ENCRYPTED !!!", "Your personal files are now scrambled.
Send 0.1 BTC to recover access.")

  # Malicious Program Popup
```

```
    messagebox.showwarning("Warning", "Backdoor established. System logs are being uploaded...")


    # Open 9gag.com
    webbrowser.open("https://9gag.com/")
    root.destroy()
```

Step5: The Logout Sequence

```
# --- 4. System logout ---


#def final_countdown():
    # Wait for 30 seconds as required
    time.sleep(30)
    # Force logout for Kali Linux
    os.system("pkilll -u $(whoami)")
```

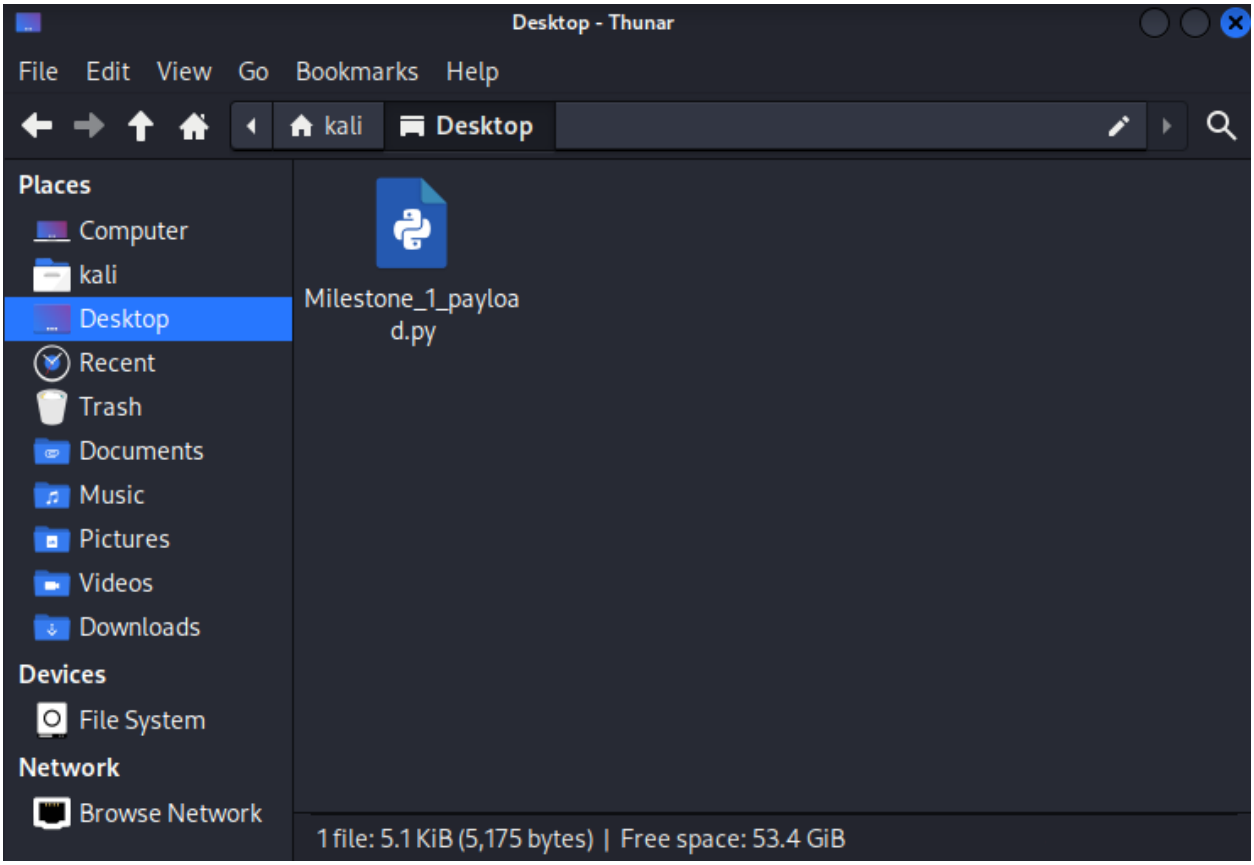
Step6: Execution

The script must call all functions in order to make it One-Click.

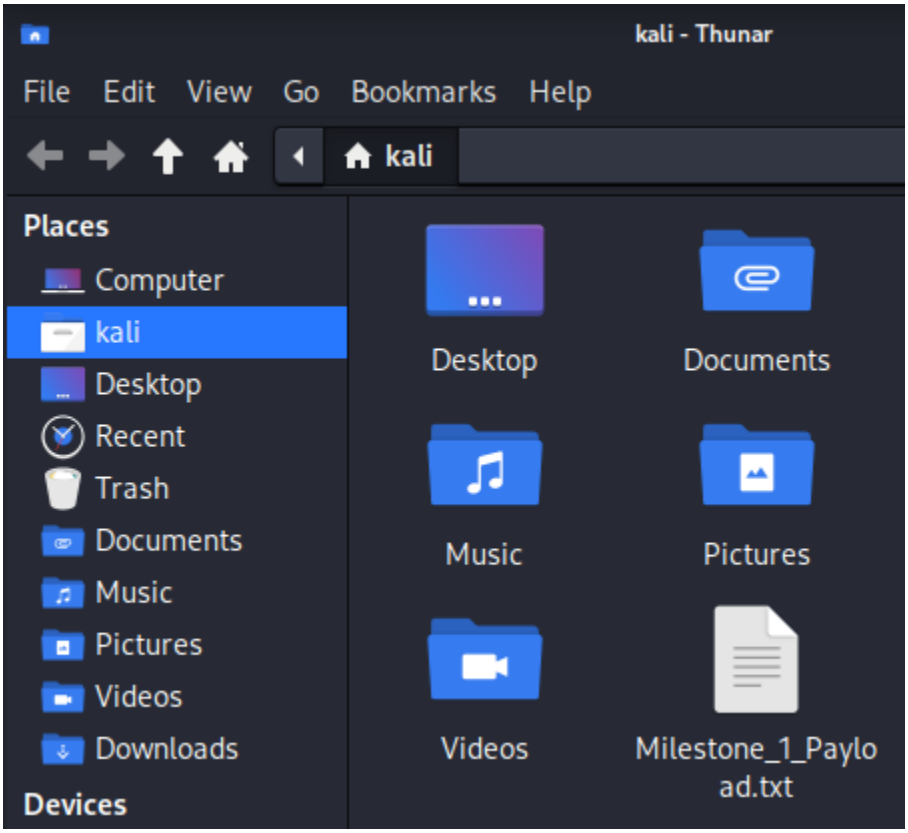
```
#--- Main Execution ---


#if __name__ == "__main__":
    create_target_file()
    encrypt_file()
    show_warnings()       # This function opens the popups
   # final_countdown()  # This function waits 30s THEN logs out
```

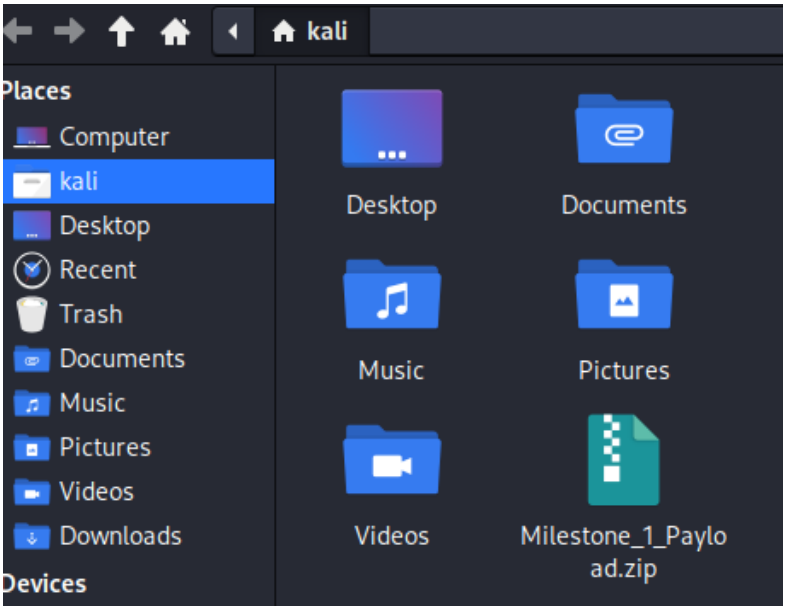
Proof of the creation of the Milestone_1_payload.py script.



Proof of the creation of the creation of the Lorem Ipsum text file that was located in the below zipped folder.



This is the zipped folder that has the Lorem Ipsum text file.



Step 2: Making the program one-click

This command, **chmod +x ~/Desktop/payload.py ,** makes the program executable.

```
┌──(kali㉿kali)-[~]
└─$ chmod -x ~/Desktop/Milestone_1_payload.py
```

The explanation of the code:

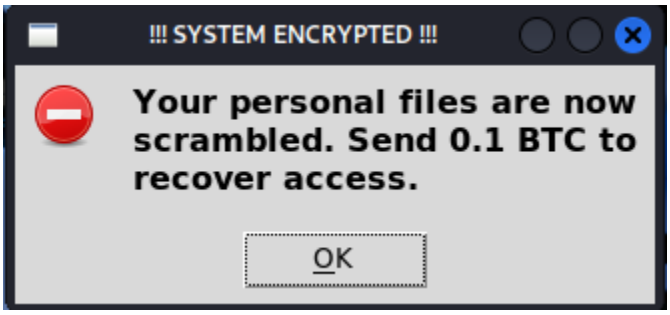
chmod +x : this line of code allows the program to run after one-click.

~/Desktop/payload.py : the location of the script.

Step 3: Executing the program

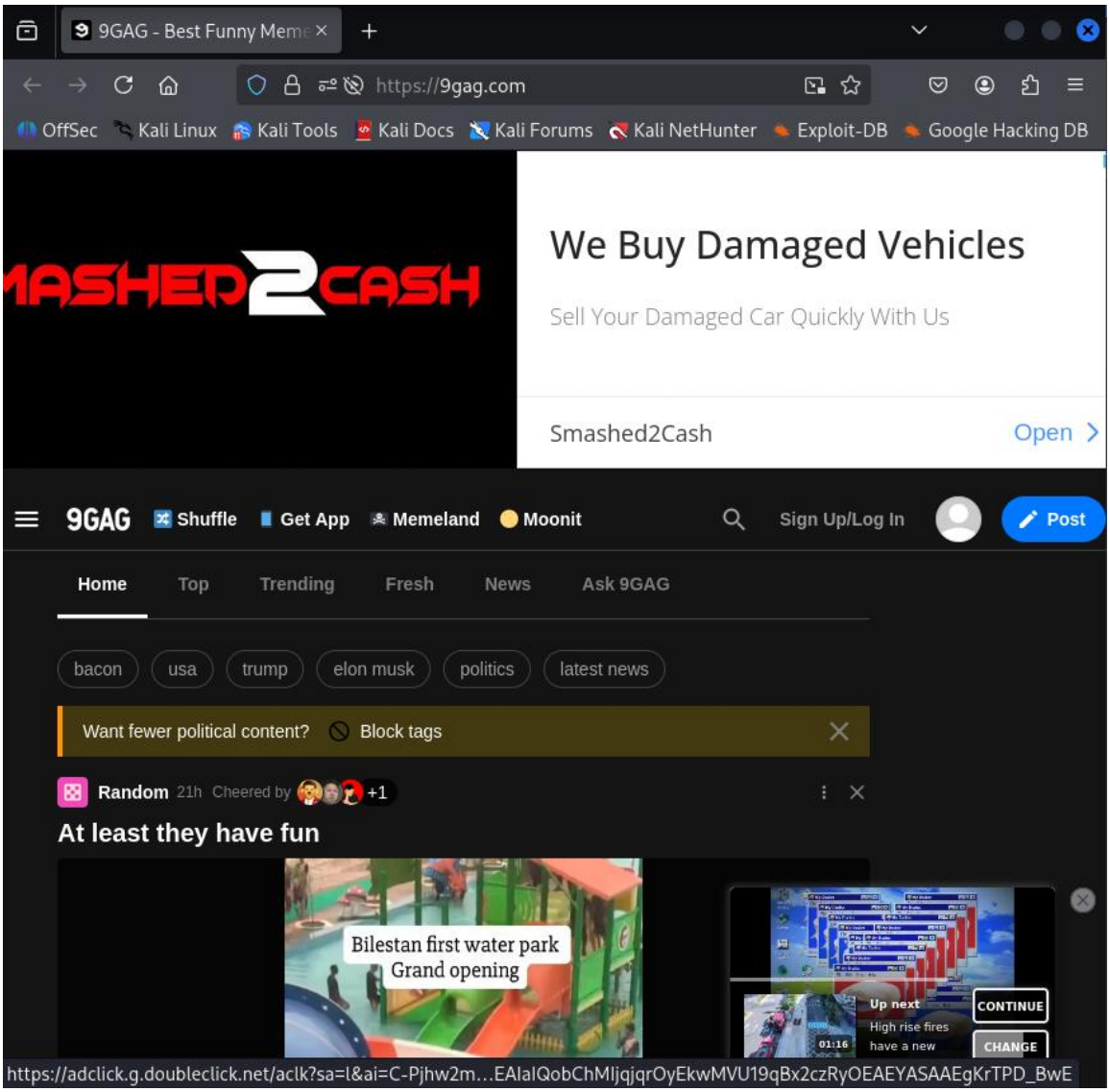
This is the Milestone_1_payload.py that can be run by clicking once.

This happens when you click on the payload. This pop-up is proof that the system is encrypted.



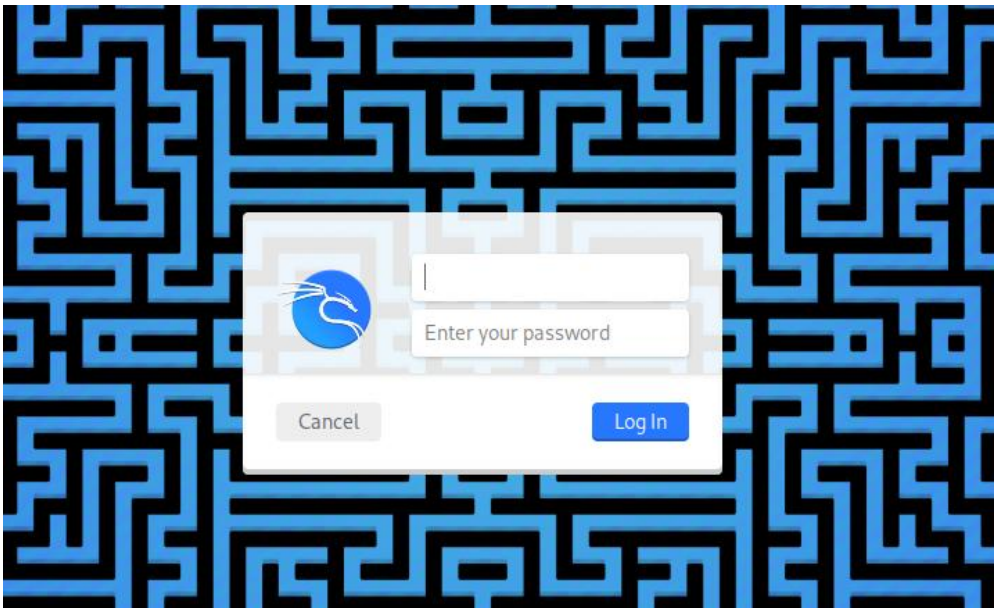
After clicking ok on the above pop-up this warning appears further displaying that there is a backdoor. When ok is clicked you will be redirected to a website called 9gag.

This is the website you will be redirected to followed by a shutdown after 30 seconds and a blank screen.



Result of the forced log out after 30 seconds.

After the shutdown you will be redirected to the login screen. This is a result of a successful ransomware attack. Log back in using kali as username and kali as password.



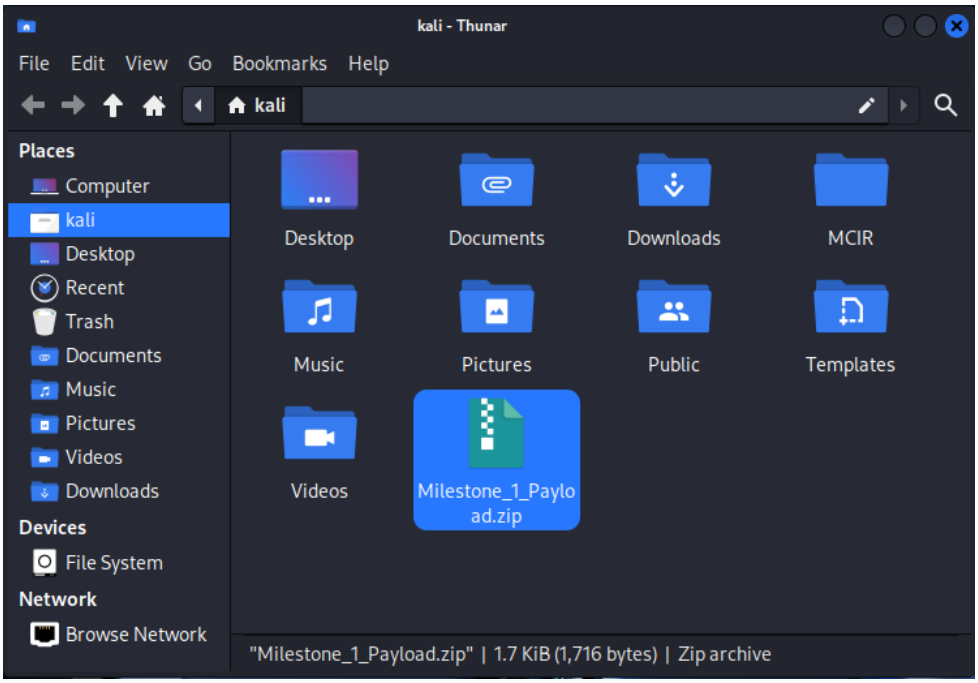
Step 4: Decryption Process

After the forced log out you can log back in using kali as the username and kali as the password

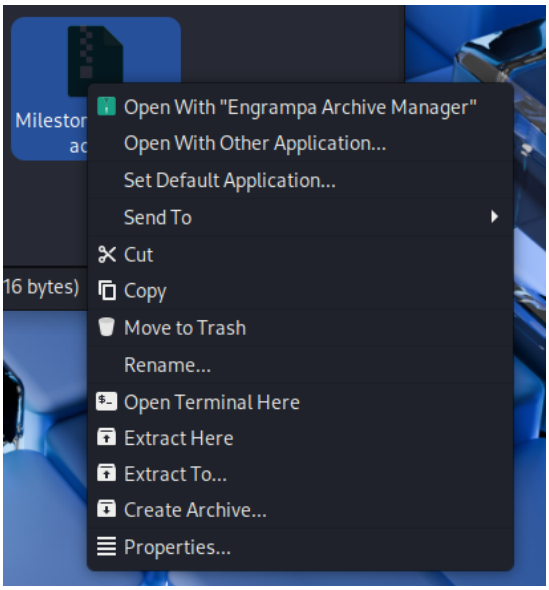
To decrypt the file , the user must extract the text file using the password '**Wednesd@y2025**' and then run the reversal script again, which will restore the original file. This is possible because the scrambling uses a string reversal method.

To do so you would need to first locate the zipped Milestone_1_payload.py and extract the Lorem Ipsum text file.

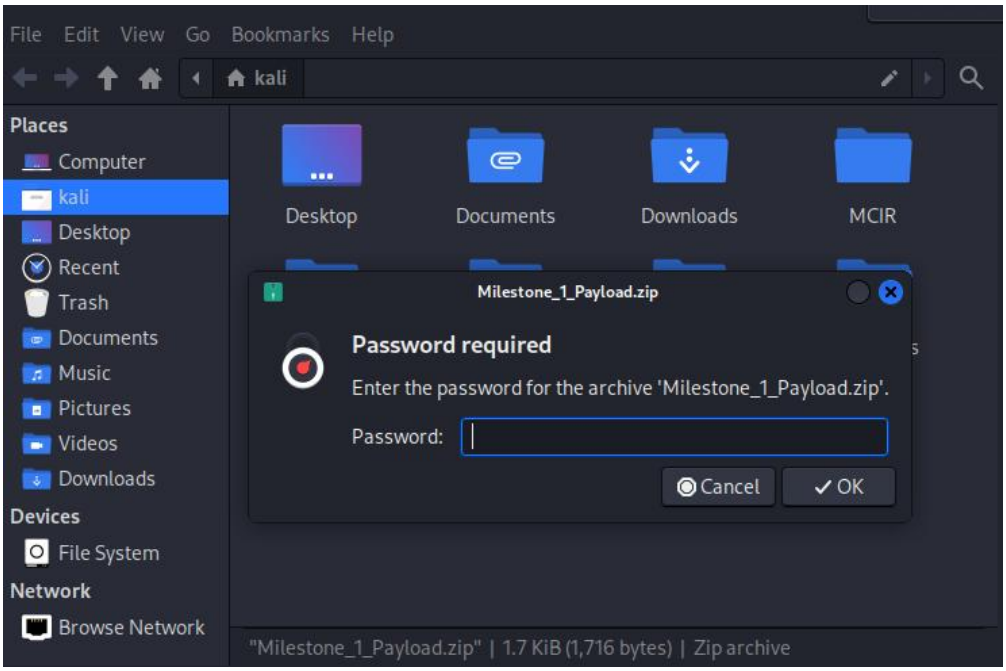
The Milestone_1_payload.zip folder is located in the kali directory .The next step is to extrate the file.



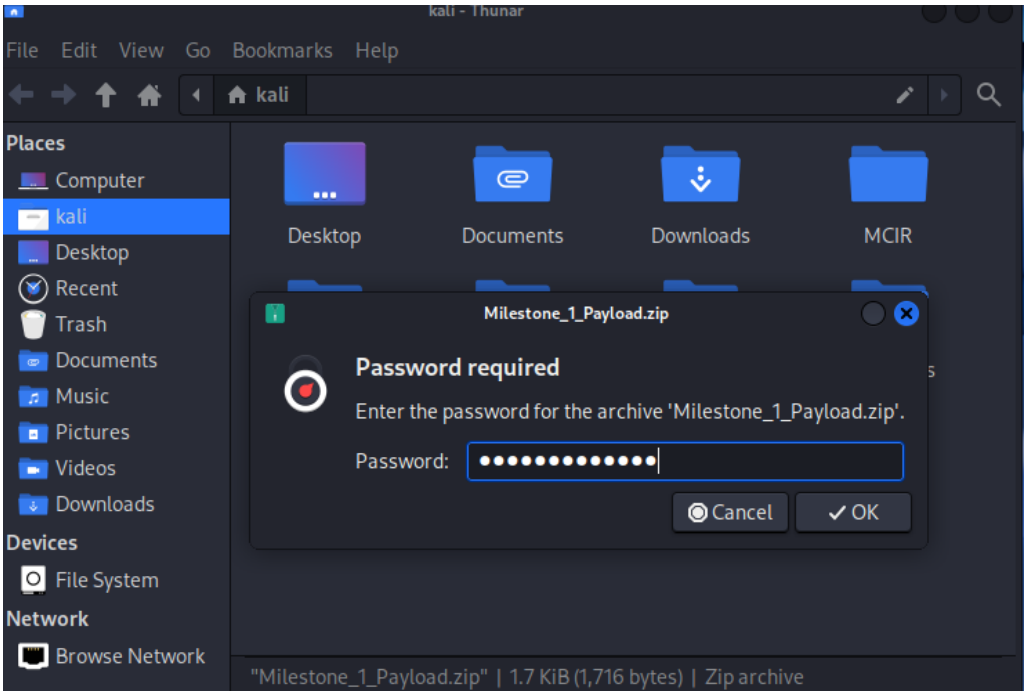
Here the folder is in the process of extracting the file.



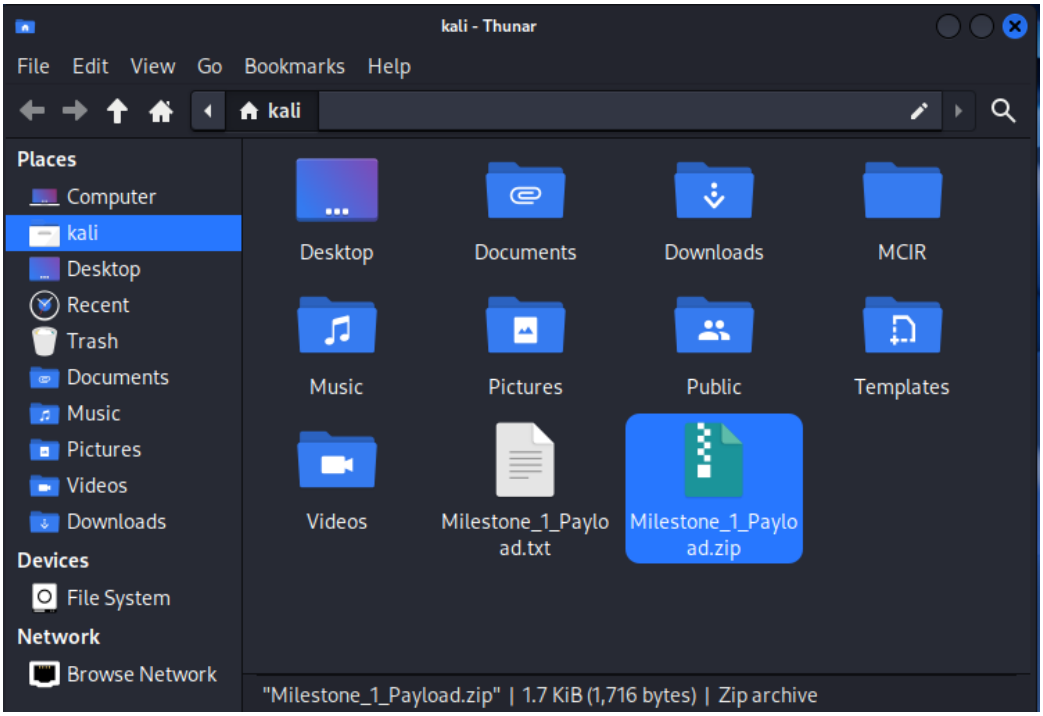
A password is required because it is the decryption key,Wednesd@y2025



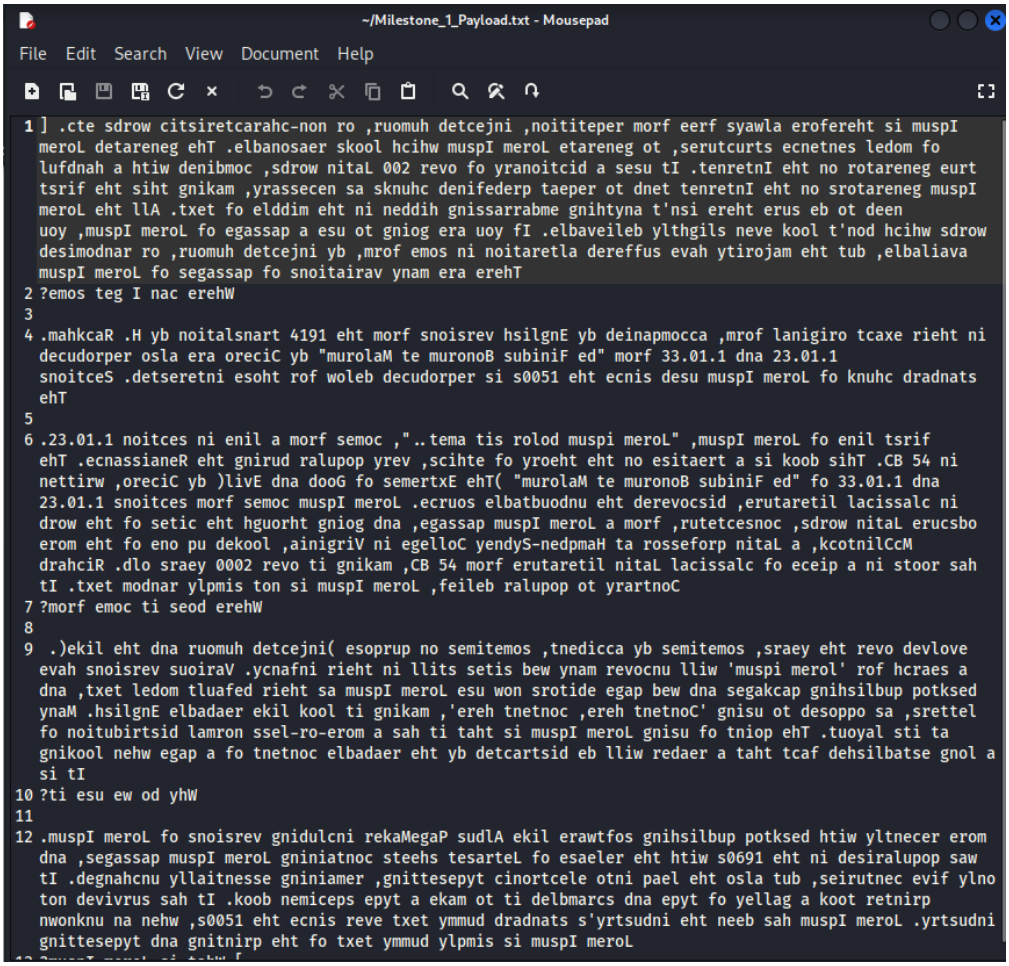
Here the password, Wednesd@y2025 is being entered to retrieve the the scrambled text file.



Result after extraction of the text file located in the kali directory.



The inside of the text file is scrambled because of use of the self-inverse string reversal method.




```
1 ] .cte sdrow citsiretcarahc-non ro ,ruomuh detcejni ,noititeper morf eerf syawla eroferehт si muspI
  meroL detareneg ehT .elbanosaer skool hcihw muspI meroL etareneg ot ,serutcurts ecnetnes ledom fo
  lufdnah a htiw denibmoc ,sdrow nitaL 002 revo fo yranoitcid a sesu tI .tenretnI eht no rotareneg eurt
  tsrif eht siht gnikam ,yrassecen sa sknuhc denifederp taeper ot dnet tenretnI eht no srotareneg muspI
  meroL eht llA .txet fo elddim eht ni neddih gnissarrabme gnihtyna t'nsi ereht erus eb ot deen
  uoy ,muspI meroL fo egassap a esu ot gniog era uoy fI .elbaveileb ylthgils neve kool t'nod hcihw sdrow
  desimodnar ro ,ruomuh detcejni yb ,mrof emos ni noitaretla dereffus evah ytirojam eht tub ,elbaliava
  muspI meroL fo segassap fo snoitairav ynam era erehT
2 ?emos teg I nac erehW
3
4 .mahkcaR .H yb noitalsnart 4191 eht morf snoisrev hsilgnE yb deinapmocca ,mrof lanigiro tcaxe rieht ni
  decudorper osla era oreciC yb "murolaM te muronoB subiniF ed" morf 33.01.1 dna 23.01.1
  snoitceS .detseretni esoht rof woleb decudorper si s0051 eht ecnis desu muspI meroL fo knuhc dradnats
  ehT
5
6 .23.01.1 noitces ni enil a morf semoc ,"..tema tis rolod muspi meroL" ,muspI meroL fo enil tsrif
  ehT .ecnassianeR eht gnirud ralupop yrev ,scihte fo yroeht eht no esitaert a si koob sihT .CB 54 ni
  nettirw ,oreciC yb )livE dna dooG fo semertxE ehT( "murolaM te muronoB subiniF ed" fo 33.01.1 dna
  23.01.1 snoitces morf semoc muspI meroL .ecruos elbatbuodnu eht derevocsid ,erutaretil lacissalc ni
  drow eht fo setic eht hguorht gniog dna ,egassap muspI meroL a morf ,rutetcesnoc ,sdrow nitaL erucsbo
  erom eht fo eno pu dekool ,ainigriV ni egelloC yendyS-nedpmaH ta rosseforp nitaL a ,kcotnilCcM
  drahciR .dlo sraey 0002 revo ti gnikam ,CB 54 morf erutaretil nitaL lacissalc fo eceip a ni stoor sah
  tI .txet modnar ylpmis ton si muspI meroL ,feileb ralupop ot yrartnoC
7 ?morf emoc ti seod erehW
8
9  .)ekil eht dna ruomuh detcejni( esoprup no semitemos ,tnedicca yb semitemos ,sraey eht revo devlove
  evah snoisrev suoiraV .ycnafni rieht ni llits setis bew ynam revocnu lliw 'muspi merol' rof hcraes a
  dna ,txet ledom tluafed rieht sa muspI meroL esu won srotide egap bew dna segakcap gnihsilbup potksed
  ynaM .hsilgnE elbadaer ekil kool ti gnikam ,'ereh tnetnoc ,ereh tnetnoC' gnisu ot desoppo sa ,srettel
  fo noitubirtsid lamron ssel-ro-erom a sah ti taht si muspI meroL gnisu fo tniop ehT .tuoyal sti ta
  gnikool nehw egap a fo tnetnoc elbadaer eht yb detcartsid eb lliw redaer a taht tcaf dehsilbatse gnol a
  si tI
10 ?ti esu ew od yhW
11
12 .muspI meroL fo snoisrev gnidulcni rekaMegaP sudlA ekil erawtfos gnihsilbup potksed htiw yltnecer erom
  dna ,segassap muspI meroL gniniatnoc steehs tesarteL fo esaeler eht htiw s0691 eht ni desiralupop saw
  tI .degnahcnu yllaitnesse gniniamer ,gnittesepyt cinortcele otni pael eht osla tub ,seirutnec evif ylno
  ton devivrus sah tI .koob nemiceps epyt a ekam ot ti delbmarcs dna epyt fo yellag a koot retnirp
  nwonknu na nehw ,s0051 eht ecnis reve txet ymmud dradnats s'yrtsudni eht neeb sah muspI meroL .yrtsudni
  gnittesepyt dna gnitnirp eht fo txet ymmud ylpmis si muspI meroL
```

Explanation of the reversal method:

To reversing the scramble the user would need to apply the same logic using the reverse slice method **([:: -1])** which restores it to its original form. Which means running the same script on the file again will simply switch the text back to its original form.

Below is the code that would be used for reversal:

```
with open("Milestone_1_Payload.txt", "r") as f:

    scrambled_data = f.read()
```

```
# Applying the reverse operation again to decrypt
original_data = scrambled_data[::-1]


with open("Decrypted_Payload.txt", "w") as f:
    f.write(original_data)
```

This code cannot be put in the main payload.py script because then it would no longer be a malicious program but rather a contradiction as it would add the solution into the attack making it self-healing.

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